Save Functional-Type Classification for One Continental Unit

Description

Saves a pre-computed functional-type classification tibble as a dated '.qs' file in 'path_processed'. The classification is produced upstream by 'assign_functional_type_clusters()' via the 'data_functional_type_classification_continent' pipeline target.

Usage

```
save_continental_functional_type_classification(
  continent_id,
  data_functional_type_classification,
  path_classification_directory = here::here("Data/Processed/Traits"),
  classification_source_prefix = NULL,
  trait_variant_id = "all_six",
  verbose = TRUE
)
```

Arguments

`continent_id`	A single non-empty character string identifying the continental unit (e.g. '"europe"', '"america"', '"asia"'). Used only for naming the output file.
`data_functional_type_classification`	A data frame containing the functional-type classification, as returned in the 'classification' element of 'assign_functional_type_clusters()'. Typically has columns 'taxon_name', 'functional_type', and 'silhouette_width'.
`path_classification_directory`	A single character string giving the directory where the '.qs' output file will be written. Default: 'here::here("Data/Processed/Traits")'.
`classification_source_prefix`	Optional single non-empty character string to prepend to the 'continent_id' portion of the file name. Use 'NULL' (default) for the historical paleo/global naming convention, or '"modern"' for modern-data functional-type classifications.

`trait_variant_id`	Functional-type trait variant identifier. The primary "all_six" variant retains the historical file name. Sensitivity variants receive a suffix.
`verbose`	Logical. If 'TRUE' (default), progress messages are printed to the console via 'cli'.

Details

Steps performed:

1. Validate arguments.
2. Save 'data_functional_type_classification' with 'RUtilpol::save_latest_file()' as a dated '.qs' file in 'path_classification_directory'. The file is only overwritten when the content has changed since the previous version.
3. Resolve the path of the just-saved file via 'RUtilpol::get_latest_file_name()'.
4. Return the file path as a character string.

Data filtering, distance computation, hierarchical clustering, ft-groups selection are all handled upstream by 'select_continental_trait_table()', 'compute_trait_dissimilarity()', 'fit_hierarchical_clustering()', and 'assign_functional_type_clusters()' respectively.

Value

A single character string: the absolute path to the '.qs' file that was written. The file is saved via 'RUtilpol::save_latest_file()' using the name stem 'data_functional_type_classification_continent_id' when 'classification_source_prefix = NULL', or 'data_functional_type_classification_classification_source_prefix_continent_id' otherwise. The final file name includes a date stamp and content hash appended by 'RUtilpol' (e.g. 'data_functional_type_classification_europe_2026-05-07__abc123__.qs').

See Also

[select_continental_trait_table()], [compute_trait_dissimilarity()], [fit_hierarchical_clustering()], [assign_functional_type_clusters()]